

Microscopic Traffic Simulation: Calibration and Dynamic applications

Task 1: Model Calibration

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Overview

1. Simulation replications
2. Measure of Performance
3. Input space exploration
4. Hyperparameter optimization
5. Early SPSA results

SIMULATION REPLICATIONS

On simulation stochasticity

Microscopic simulations with SUMO are inherently stochastic: the simulator uses RNGs with different (configurable) seeds to simulate random human behavior (rerouting probability, driving styles, route choice).

This reflects in the results of the simulation themselves: a simulation run with the same exact parameters will yield different results based on the RNG seed.

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How do we deal with this?

Required simulation number

We **average** multiple simulation results. The number of simulations required depends on the tolerance and confidence level we want to achieve.

$$n = \left(\frac{t_{\alpha/2, n-1} \cdot \sigma}{W} \right)^2$$

where:

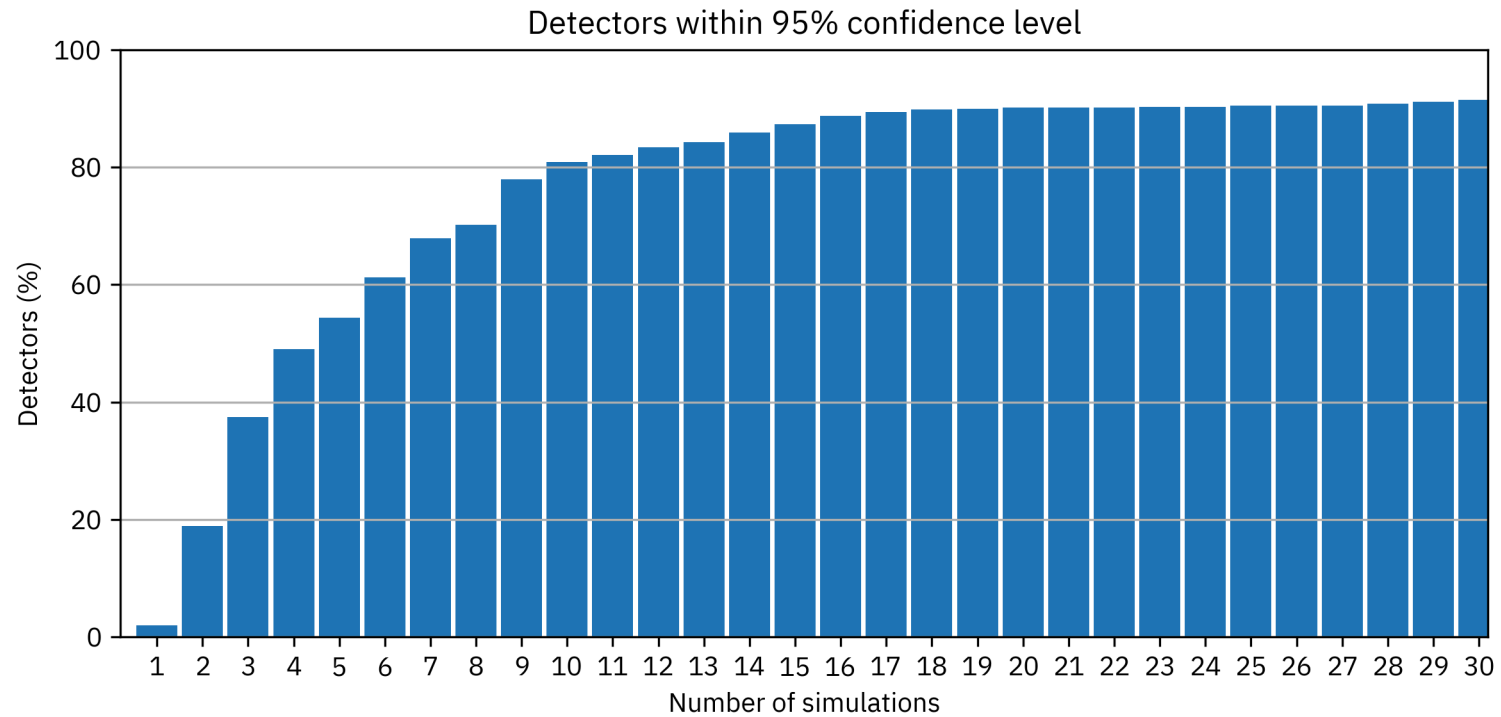
n number of required simulations

t t-students statistic (for $\alpha/2$ significance, $n - 1$ DOF)

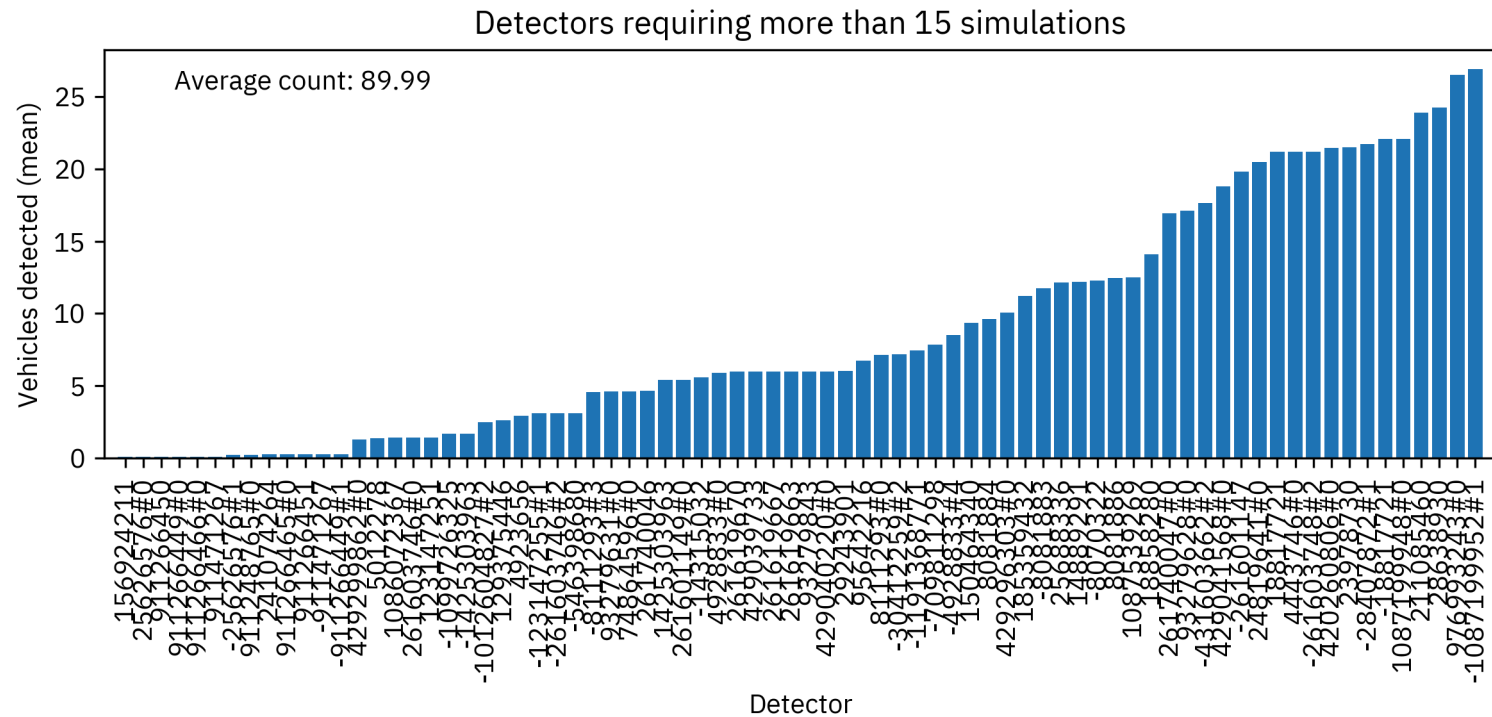
σ standard deviation

W width of tolerance interval

Required simulation number



Excluded detectors



Simulation replications

MEASURE OF PERFORMANCE

Limits of common objective functions

Some common objective functions, like MAPE and RMSE, show their limits when applied to a traffic calibration problem.

RMSE

- Absolute metric: can put errors of different significance on the same scale (e.g. 100 vehicles on an Autobahn / Residential street)
- Dominated by outliers (as errors scale quadratically)

MAPE

- Unstable on small flows: when the denominator is small, MAPE can explode
- Asymmetric

Benefits of RMSN

- **Independent from scale:** allows to compare GOF across networks / traffic conditions
- **Normalization:** weights the RMSE against the global mean of observations, avoiding both instability on small flows and outlier dominance
- **Interpretability:** can be interpreted as an estimate of the coefficient of variation and expressed as a percentage

GLS-based weighting schemes

Generalized least squares weight errors based on the variance of observations, using the inverse of the variance-covariance matrix V :

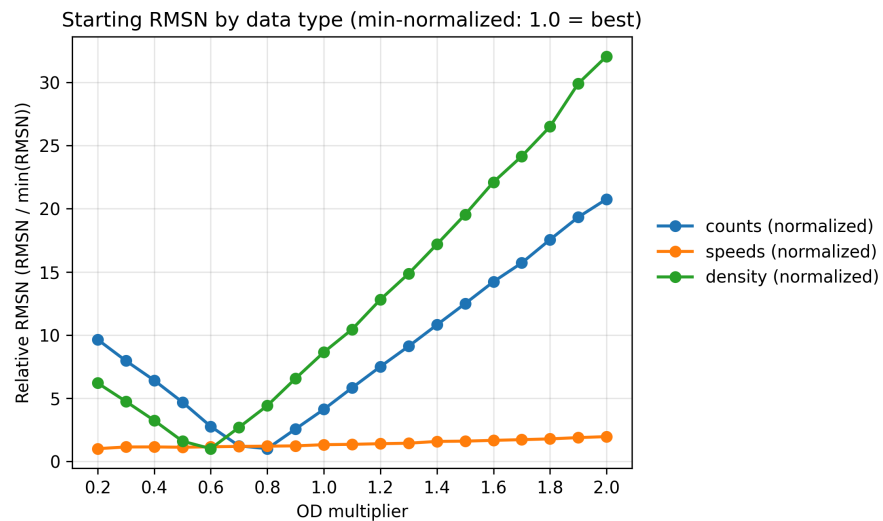
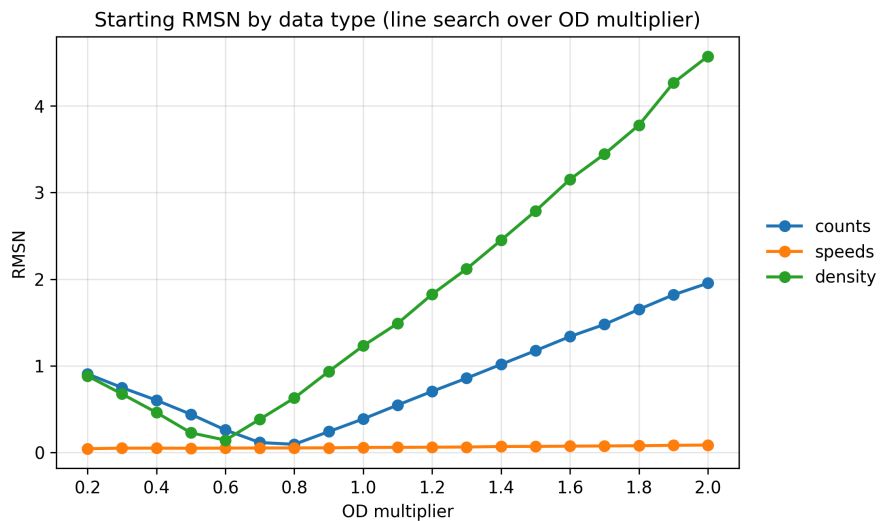
$$\text{GLS} = (Y_{\text{sim}} - Y_{\text{true}})^T V^{-1} (Y_{\text{sim}} - Y_{\text{true}})$$

The use of the matrix V allows to account for both heteroscedasticity and spatial correlation of observations.

However, in the absence of an a priori information on V , we would need to estimate it from data.

INPUT SPACE EXPLORATION

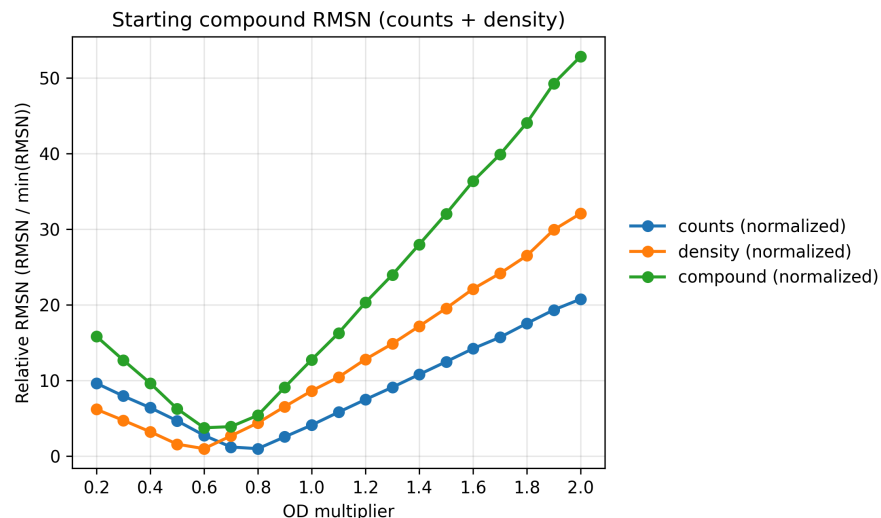
Line search on starting RMSN



Line search on starting RMSN

As the speed RMSN seems to not be particularly affected by changes in the OD structure, it was given a very low weight in the final compound measure.

$$w_c, w_d \in (0.3, 0.6), w_s = 0.1$$



HYPERPARAMETER OPTIMIZATION

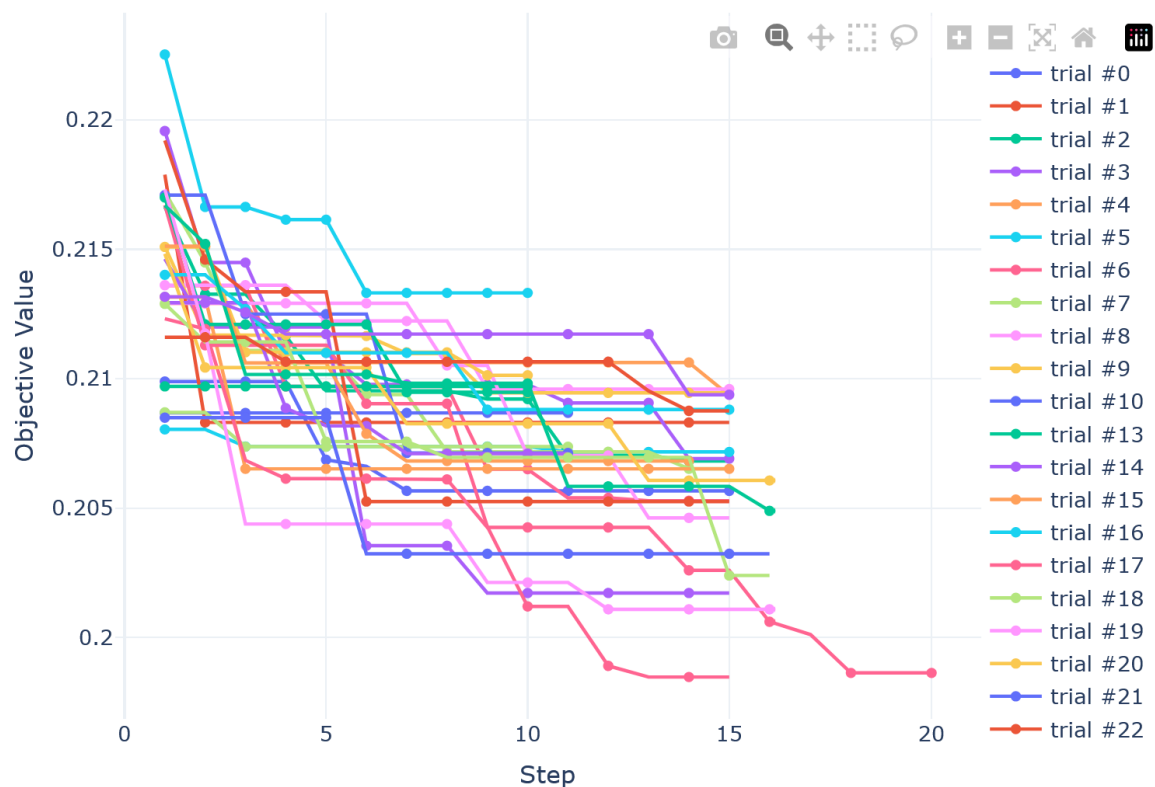
Hyperparameter optimization with OPTUNA

Three SPSA parameters where optimized using the OPTUNA library:

- Initial step size a
- Perturbation size c
- Stability constant A

OPTUNA allows to perform Bayesian optimization of a model hyperparameters, which leads to a much more efficient exploration of the parameter space compared to grid or random search.

Results of hyperparameter optimization

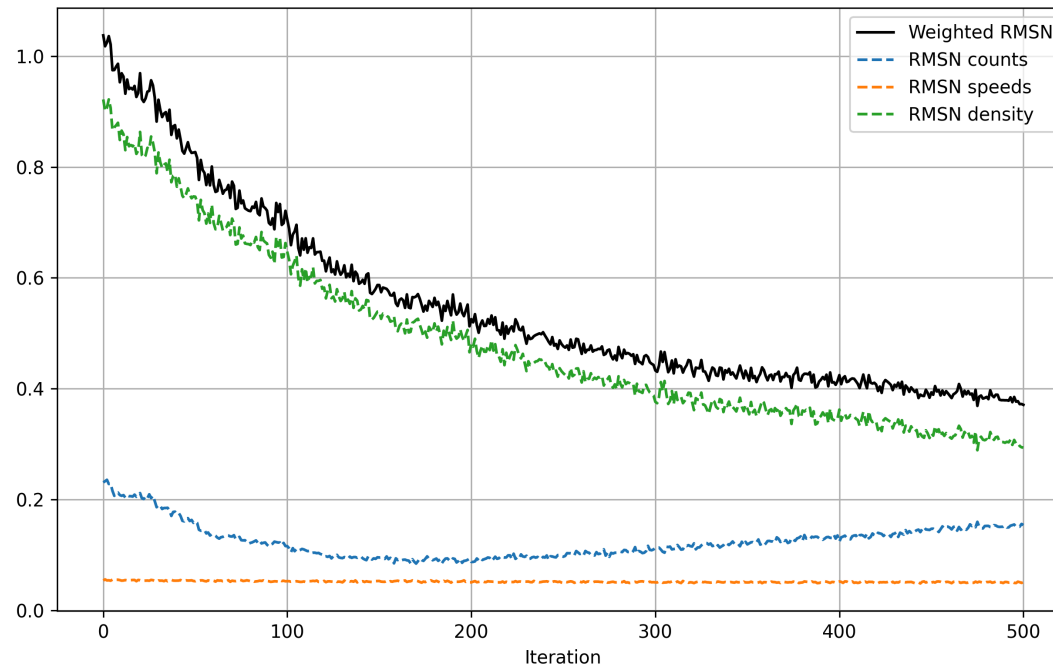


Final parameter selection:

- $a = 100$
- $c = 0.6$
- $A = 40$

EARLY SPSA RESULTS

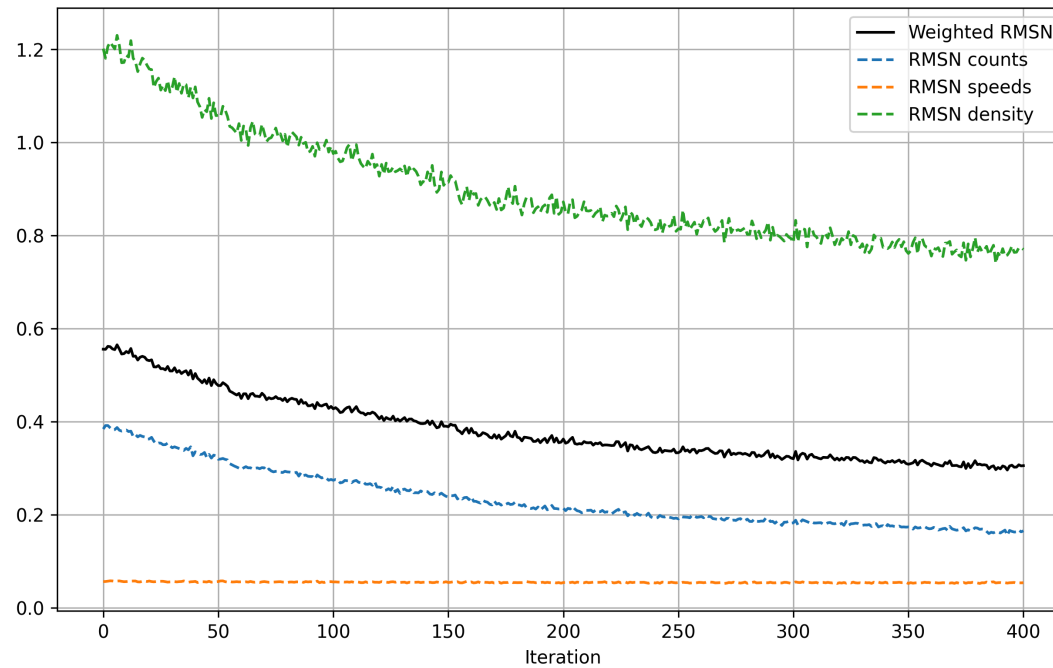
RMSN components progression



Weights:

- $w_c = 0.3$
- $w_d = 0.6$
- $w_s = 0.0$

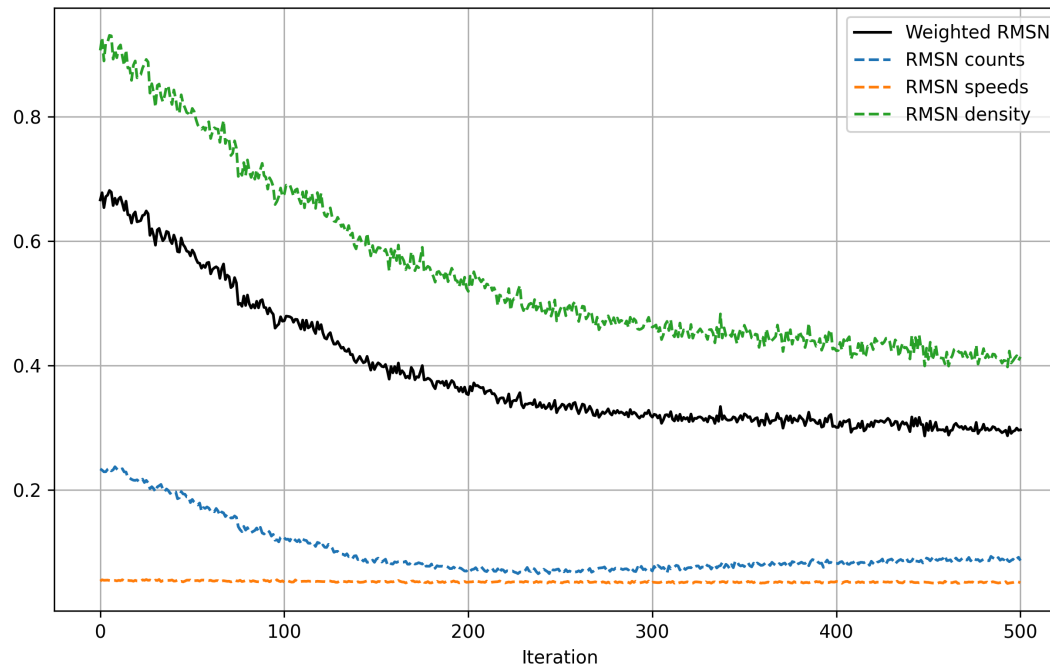
RMSN components progression



Weights:

- $w_c = 0.5$
- $w_d = 0.2$
- $w_s = 0.1$

RMSN components progression

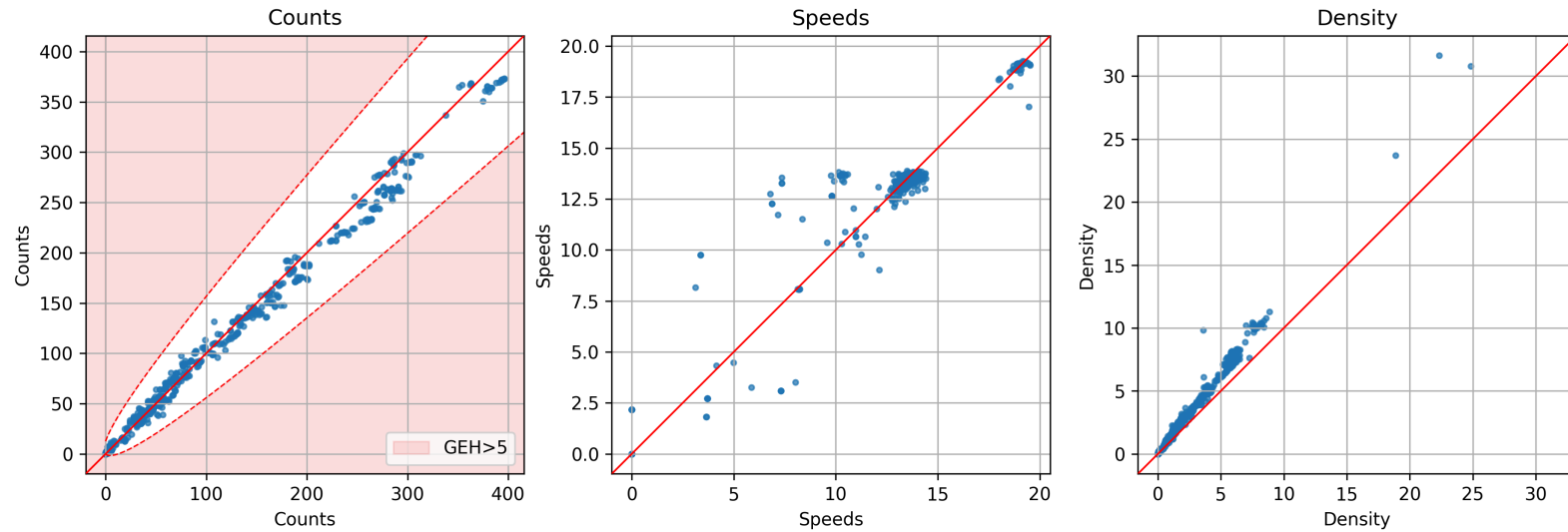


Weights:

- $w_c = 0.5$
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Predicted vs Actual flows

Predicted vs Actual (best RMSN at iteration 448)



Thank you!